

Oct-22/TE/Insem-557

T.E. (Information Technology)

THEORY OF COMPUTATION

(2019 Pattern) (Semester - I) (314441)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates :

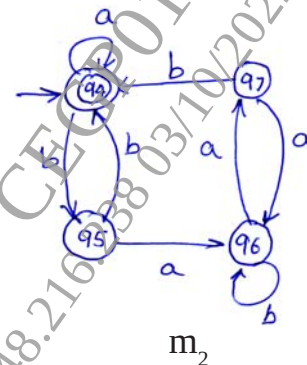
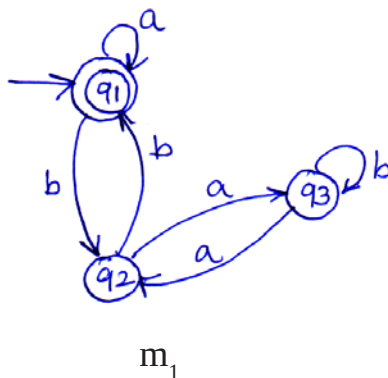
- 1) Answer Q.1 or Q.2, Q.3 or Q.4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) Design a DFA which accepts a binary number divisible by 4. [5]
 b) Design a Mealy machine to increment binary number by 1. Write down transition table. [4]
 c) Convert the following NFA with ϵ -moves to DFA. [6]

State/input	δ			
	ϵ	a	b	c
$\rightarrow p$	{q}	{p}	ϕ	ϕ
q	{r}	ϕ	{q}	ϕ
r^x	ϕ	ϕ	ϕ	{r}

OR

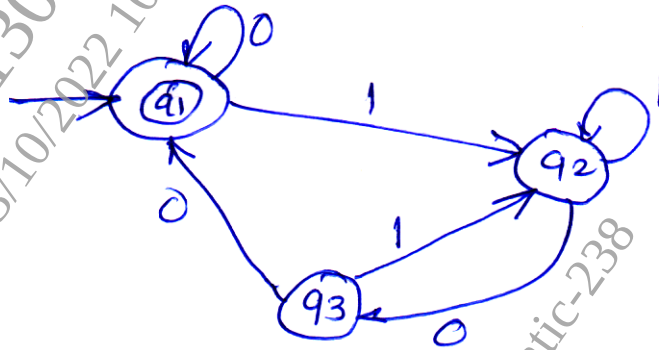
- Q2)** a) Define the following terms with proper examples. [6]
 i) Alphabets
 ii) String
 iii) Natural language
 b) Show whether the following automata m_1 & m_2 are equivalent or not. [5]



P.T.O.

- c) Construct a DFA over the alphabet $\{a, b\}$ for accepting the strings ending with "ab". [4]

- Q3) a)** Find the regular expression for the set of strings recognized by the given FA using Arden's theorem. [5]



- b) Determine the regular expression over the alphabet $\{0, 1\}$ for the following : [6]

- All the string containing exactly two 0's
- All the string that do not end with 01
- All the string containing 1 as a third character from end.

- c) Explain the following terms : [4]

- Kleene closure
- Positive closure

OR

- Q4) a)** Explain any three closure properties of Regular language. [6]

- b) What is a Regular expression? Explain in brief the applications of regular expressions. [5]

- c) Construct a NFA for the following RE using direct method [4]

RE = $(ab + ba)^*aa$

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Total No. of Questions : 4]

SEAT No. :

P-5054

[Total No. of Pages : 3

[6187]-457

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(2019 Pattern) (Semester - I) (314441)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) *Attempt Q.1 or Q.2, Q.3 or Q.4.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Assume suitable data, if necessary.*

Q1) a) Design a Mealy machine to find 2's complement of any binary number.
Write the definition of a Mealy Machine. **[4]**

b) Find the final DFA by performing the DFA minimization process. A is initial state and final states are B, C, D, E, F, G. **[5]**

Q/Σ	0	1
A	B	-
B	C	C
C	D	E
D	C	C
E	B	G
F	E	-
G	B	G

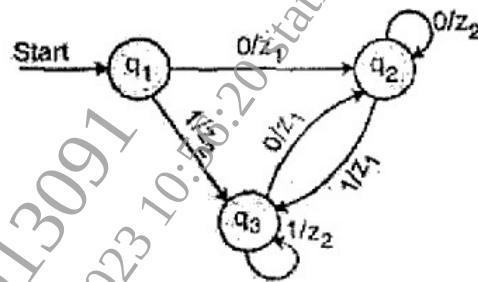
c) Write the formal definitions for the following : **[6]**

- i) NFA (Non-Deterministic Finite Automata)
- ii) Mealy Machine
- iii) Moore Machine

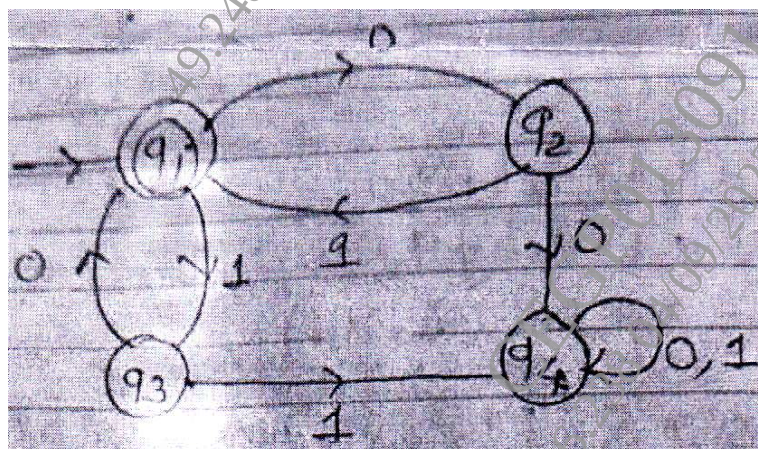
OR

P.T.O.

- Q2) a)** Construct a Moore machine equivalent to the Mealy machine represented by the following Transition Diagram : [5]



- b) Design a DFA (Deterministic Finite Automata) that reads strings made up of the letters in the word 'UNIVERSITY' and recognizes those strings that contain the word 'UNITY' as a substring. [5]
- c) Justify that there can be the equivalent Mealy machine for any Moore machine by suitable example. [5]
- Q3) a)** For the following regular expressions, draw the FA (Finite Automata) recognizing the corresponding language. [5]
- $1(01 + 10)^* + 0(11 + 10)^*$
 - $(01 + 10)^* 00 (01 + 11)^*$
- b) State and explain Pumping Lemma for a regular language. [5]
- c) Find the regular Expression for the FA (Finite Automata) using Arden's Theorem. [5]



OR

- Q4)** a) Use pumping lemma to check whether the language, $L = \{0^n \mid n \text{ is prime}\}$ is regular or not. [5]
- b) Find the regular expression for the language over inputs $\{a, b\}$. [5]
- i) The set of all strings ending neither in 'b' nor in 'ba'
 - ii) The set of all strings containing at least one 'a' and at least one 'b'
- c) Prove that "Regular language is closed under Kleene Star". [5]



Total No. of Questions : 4]

SEAT No. :

PE53

[Total No. of Pages : 2

[6579]-485

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THEORY OF COMPUTATION

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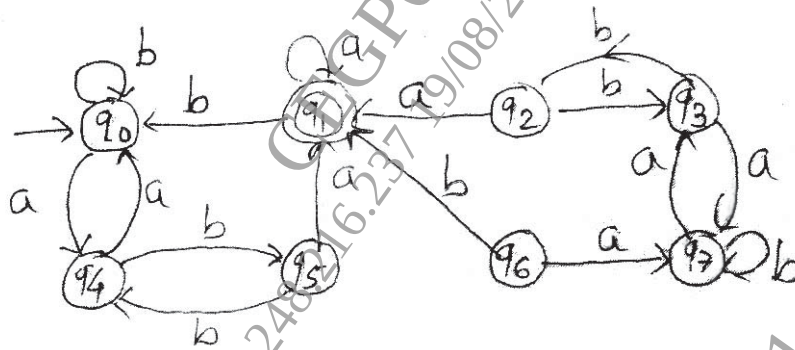
Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Attempt Q1 or Q2, Q3 or Q4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) Construct a DFA (Deterministic Finite Automata) which accepts on octal number divisible by 3. [5]
- b) Construct the minimum state automata equivalent to the following DFA (Deterministic Finite Automata). [5]

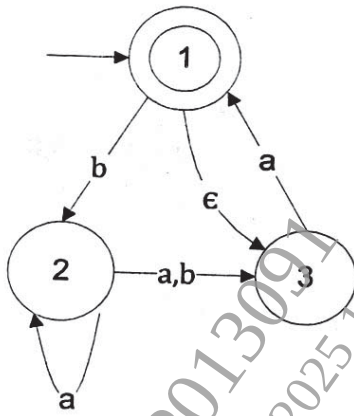


- c) Design a Mealy Machine accepting the language of strings From Σ^* where $\Sigma = \{0,1\}$ and ending with 00 or 11. [5]

OR

- Q2)** a) Construct a DFA (Deterministic Finite Automata) over the language $\Sigma = \{0,1,2\}$ to accept all the strings having substring '120'. [5]
- b) Construct DFA (Deterministic Finite Automata) for the following NFA (Non-Deterministic Finite Automata). [5]

P.T.O.



c) Compare Moore Machine and Mealy Machine. [5]

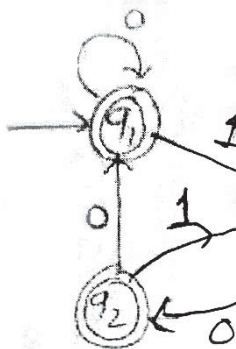
Q3) a) For the following regular expressions, draw the FA (Finite Automata) recognizing the corresponding language. [5]

i) $(0+1)^*(1+00)(0+1)^*(1+01)$

ii) $0(1+10)+10^*+01^*0$

b) State and explain Pumping Lemma for a regular language. [5]

c) Find the regular expression for the FA (Finite Automata) using Arden's Theorem. [5]



OR

Q4) a) Use pumping lemma to check whether the language, $L = \{ww \mid w \in \{0,1\}^*\}$ is regular or not. [5]

b) Find the regular expression for the language over inputs $\{0,1\}$ [5]

i) All string containing 011 at the end.

ii) String contains atleast two zeros.

c) Prove that "Regular language is closed under union and concatenation". [5]

